Lever Test · Teacher and Cover Plan

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Scheme Outcome

TRACK Science scheme of work, Unit 1 (Forces, Mechanisms and Electricity), Topic: Investigate how changing the distance from the pivot affects the effort needed to lift a load.

Carry out a systematic lever investigation changing effort distance from the pivot, record the effort needed, and state the pattern from the evidence.

Success Criteria

I can keep the load and pivot position the same while changing the effort distance.

I can record the effort needed (in coins/masses or newtons) at three distances.

I can use my results to state how effort distance affects the force needed.

Model Sequence (I do)

First, check the controls: the pivot stays fixed and the same load sits 5 cm from the pivot every time.

Second, compare the three effort distances: at 5 cm it took 8 coins, at 10 cm it took 4 coins, and at 20 cm it took 2 coins.

Third, state the pattern: as the distance from the pivot to the effort increases, the effort needed

decreases.

If our coin stack slips off the mark, the effort distance changes and gives a faulty reading.

So we align the centre of the coin stack with the exact ruler mark each time.

If a result looks surprising, we keep it in the table, check the ruler mark, and repeat the trial.

We Do & Hinge Check

At 10 cm from the pivot, lifting the load took 4 coins. At 20 cm it took 2 coins. Roughly how many coins would it take at 15 cm from the pivot?

Answer: A. Using the pattern, 15 cm lies between 10 cm and 20 cm, so the effort needed is between 4 coins and 2 coins (about 3 coins).

Safe Practical & Card Setup

Test three effort distances (5 cm, 10 cm, 20 cm from the pivot) on the ruler lever or use the supplied results.

Kit: 30 cm stiff ruler, fixed triangular pivot block, load mass taped at 5 cm from pivot, and identical 2p coins.

Keep the lever flat on the middle of the desk. Stack coins gently; do not flick the ruler.

You Do Routes (Supported, Independent, Aspire)

Supported: Complete the coin counts for 5 cm, 10 cm, and 20 cm and circle which distance needed the least

effort. (Help: 5 cm = 8 coins. 10 cm = 4 coins. 20 cm = 2 coins. 20 cm needed the least effort.)

Independent: Record the results for 5 cm (8 coins), 10 cm (4 coins), and 20 cm (2 coins), then write a conclusion quoting the numbers and naming one control variable. (Help: At 5 cm it took __ coins, while at 20 cm it took __ coins. This shows that as effort distance increases, the effort needed __.)

Aspire: Use the quantitative pattern (5 cm -> 8, 10 cm -> 4, 20 cm -> 2) to explain what happens when effort distance doubles, and state one source of error in a ruler lever. (Help: Every time the effort distance from the pivot doubles, the effort needed __. One source of error is the mass of the __ itself.)